

FORT ST JAMES, BC, Canada

WMO: 719330

Lat: 54.455N

Lon: 124.286W

Elev: 686

StdP: 93.36

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.0	-36.1	-43.5	0.1	-38.9	-40.9	0.1	-36.2	8.3	-15.5	7.5	-17.1	1.3	50	0.686

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	27.8	17.5	26.0	16.8	24.1	16.0	18.9	24.9	18.0	23.7	17.1	22.4	4.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	13.2	21.2	16.0	12.3	20.2	15.0	11.6	19.5	56.8	24.9	53.7	23.4	50.8	22.3	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.4	6.5	DB	-42.4	31.0	2.5	1.6	-44.2	32.1	-45.7	33.1	-47.0	34.0	-48.8	35.2	(4)
(5)			WB	-42.4	20.6	2.5	1.2	-44.2	21.4	-45.7	22.1	-47.1	22.8	-48.9	23.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.9	-20.5	-18.2	-11.4	-0.7	7.8	14.4	17.1	15.0	9.4	1.7	-9.3	-17.2	(6)	
	DBStd	14.82	9.22	8.02	9.05	6.80	5.37	3.93	3.15	3.65	4.40	4.67	7.00	8.69	(7)	
	HDD10.0	4587	945	789	664	323	104	9	0	6	63	261	578	843	(8)	
	HDD18.3	7065	1203	1023	923	572	327	127	61	113	270	517	828	1102	(9)	
	CDD10.0	607	0	0	0	2	37	141	222	160	44	2	0	0	(10)	
	CDD18.3	43	0	0	0	0	1	8	23	9	1	0	0	0	(11)	
	CDH23.3	549	0	0	0	0	24	123	272	120	11	0	0	0	(12)	
	CDH26.7	96	0	0	0	0	3	21	51	21	1	0	0	0	(13)	
(14) Wind	WSAvg	2.9	2.5	2.6	2.9	3.4	3.5	3.2	2.9	2.8	2.8	2.8	2.6	2.5	(14)	
(15) Precipitation	PrecAvg	481	48	28	27	21	39	46	47	48	45	43	41	49	(15)	
	PrecMax	640	126	75	60	52	107	113	124	103	97	76	77	104	(16)	
	PrecMin	348	14	5	5	5	5	7	8	6	9	19	1	12	(17)	
	PrecStd	81	26	16	13	12	23	26	28	28	23	16	16	22	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.1	3.1	9.2	18.2	26.2	29.1	30.2	29.0	24.9	16.9	7.8	2.9	(19)	
		MCWB	0.2	1.1	3.5	9.1	13.9	17.3	18.8	18.8	16.3	10.6	4.4	0.5	(20)	
	2%	DB	-1.2	-1.0	6.1	14.4	23.0	26.2	28.0	26.2	21.2	13.1	4.0	-0.2	(21)	
		MCWB	-2.4	-2.7	1.9	7.0	12.2	15.9	17.7	17.7	14.3	8.4	1.7	-1.4	(22)	
	5%	DB	-4.3	-3.9	3.9	11.8	20.1	24.3	26.2	24.2	19.0	10.6	1.7	-3.0	(23)	
		MCWB	-5.0	-5.1	0.7	5.5	11.1	15.2	17.3	16.7	13.2	6.7	0.5	-3.8	(24)	
	10%	DB	-7.9	-6.8	1.2	9.1	17.5	22.7	24.6	22.2	16.9	8.2	0.0	-5.2	(25)	
		MCWB	-8.5	-7.6	-1.0	4.2	9.9	14.7	16.8	15.8	12.1	5.7	-0.9	-5.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.4	1.2	4.1	9.6	15.2	19.0	20.4	20.4	16.9	11.5	4.6	0.7	(27)	
		MCDB	1.5	3.2	8.6	17.0	23.5	26.0	27.3	26.4	22.9	15.8	7.1	2.6	(28)	
	2%	WB	-2.4	-2.6	2.3	7.3	13.2	17.6	19.2	18.6	15.1	8.7	2.0	-1.7	(29)	
		MCDB	-1.3	-1.4	5.4	13.5	21.0	23.7	25.2	24.2	19.9	12.1	3.3	-0.7	(30)	
	5%	WB	-5.3	-5.1	0.8	5.8	11.8	16.5	18.4	17.6	13.8	7.1	0.3	-3.9	(31)	
		MCDB	-4.5	-3.9	3.5	11.1	18.9	22.2	24.1	22.7	17.6	9.8	1.3	-3.2	(32)	
	10%	WB	-8.5	-7.5	-0.9	4.3	10.4	15.4	17.5	16.5	12.8	5.8	-1.0	-6.2	(33)	
		MCDB	-8.0	-6.6	1.4	8.8	16.6	20.9	22.9	21.2	16.1	8.0	-0.1	-5.6	(34)	
(35) Mean Daily Temperature Range	MDBR		9.9	11.7	13.2	12.4	12.8	12.7	12.2	11.9	11.0	8.5	8.5	9.3	(35)	
	5% DB	MCDBR	12.6	12.9	13.5	15.3	16.9	15.4	14.8	14.8	14.2	12.0	10.0	10.5	(36)	
		MCWBR	11.6	11.3	9.9	9.3	8.3	7.4	7.0	7.6	8.1	7.9	8.1	9.5	(37)	
	5% WB	MCDBR	12.3	12.9	13.3	14.6	15.8	13.6	13.0	13.4	12.4	11.0	9.2	10.1	(38)	
		MCWBR	11.5	11.5	10.1	9.2	8.4	7.0	6.7	7.4	7.5	7.7	7.9	9.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.246	0.266	0.282	0.317	0.333	0.348	0.336	0.338	0.305	0.293	0.274	0.234	(40)	
	taud		2.331	2.331	2.341	2.400	2.476	2.454	2.525	2.515	2.610	2.594	2.388	2.321	(41)	
	Ebn at Noon		735	840	905	913	914	900	905	883	880	813	687	671	(42)	
	Edh at Noon		53	75	94	102	101	105	96	91	73	59	51	43	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.67	1.58	2.90	4.50	5.41	5.90	5.70	4.84	3.21	1.70	0.75	0.50	(44)	
	RadStd		0.08	0.19	0.23	0.22	0.40	0.48	0.52	0.23	0.28	0.19	0.10	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page